

tion. Congestion in wireless networks is of two types: radio collision and buffer overflow. Congestion is very dangerous for WSNs. Collision has the following drawbacks:

1. Increases energy dissipation rates of sensor nodes
2. Causes a lot of packet loss which in turn diminishes the network throughput
3. Hinders fair event detections and reliable data transmissions

Inter symbol interferences (ISI)

Multi-path, cross-antenna interference and multi-user interference (MUI) are the major vulnerabilities of wireless communication systems. To avoid these, a system uses multi-carrier modulation, which is very effective in avoiding MUIs and providing multiple access simultaneously.

Recent developments of orthogonal frequency division multiplexing (OFDM) and interleave division multiple access (IDMA) have proved that OFDM suppresses the inter symbol interference (ISI) induced by multi-path channels and IDMA avoids MUIs. The drawbacks of code division multiple access (CDMA) are overcome by OFDM-IDMA techniques. In CDMA, interleaver is used for coding gain while in IDMA, the interleavers are used for separating the users. The basic function of IDMA is to use specific interleaver to differentiate the users. However, it is imperative that the interleaver between the transmitter and receiver should be identical. The WSN life-time is dependent on the available energy, so energy routing protocol is desired and can significantly prolong the life-time of WSN.

Routers

Since a distributed network has multiple nodes and services many messages, and each node can be a shared resource, many decisions must be made. There may be multiple paths from the source to the destination, so the message routing is an important task. Routers are a specific type of measurement nodes that one can use to extend WSN working distance or its reliability. Here it is the necessity of

designing dedicated routing protocols which consumes resources like energy and bandwidth very effectively.

Routing methods can be fixed (that is, pre-planned), adaptive, centralised, distributed, broadcast, etc. Routing tables are provided in the fixed routing schemes, where they dictate the next node to be routed to, given the current message location and the destination node. Routing algorithms can be based on various network analysis and graph theoretic concepts in computer science (for example, A-star tree-search, or in operations research including shortest route, maximal-flow and minimum-span problems). Routing is closely associated with dynamic programming and the optimal control problem in feedback control theory

Important design issues for routing are: network model, node deployment, node/link heterogeneity (different types of data to be routed in the same network), data reporting model, energy consumption without losing accuracy, network dynamics (for example, mobility of sensor nodes is necessary, fault tolerance, connectivity, coverage, QoS, data aggregation, transmission media and security).

Resource-aware routing

Different approaches have been developed which consider sensors' resources (such as energy) when making routing decisions. These routers can be of different types, such as:

Distributed activation based on predetermined routes (DAPR). It considers the importance of a node to the sensing application. Here, a node first finds the sub-region within its region of coverage that is the most poorly covered. The cost assigned to that node is related to the combined energy of all nodes capable of redundantly covering this poorly-covered region.

Data-centering routing. In data-centering routing, the end nodes, the sensors themselves, are less important than the data. Thus, queries are posed for specific data rather than for data from a particular sensor, and routing is performed using knowledge that it

is the aggregate data rather than any individual data item that is important.

Sensor protocol for information via negotiation (SPIN)

Because of blind flooding or gossiping data dissemination, SPIN avoids three costly problems: implosion, overlap and resource blindness.

Implosions occur in highly-connected networks that employ flooding and thus each sensor receives many redundant copies of the data. For large data messages, this wastes considerable energy.

Overlap occurs if two sensors send some common data, causing redundancy in data transmission and thus energy waste. SPIN solves this problem by naming data so that sensors only request the data or parts of data they are interested in receiving

Resource blindness is a mechanism whereby a node that is running low on energy will not advertise its data in order to save its low energy resources.

To perform a sensing task, a querying node creates an interest, which is named according to the attributes of data or events to be sensed. When an interest is created, it is injected into the network by the sink node by broadcasting an interest message containing the interest type, duration and an initial reporting rate to all neighbours. Interests are diffused throughout the network toward the sink node using a forwarding technique.

Classification of WSNs

Network based. These are based on mobility, network type and type of communication. Their mobility can be static or dynamic. The network can be flat or cluster-type. The flat type is data-centric and all the nodes in the network are treated equally.

Clustering type. Different nodes are grouped to form clusters and data from nodes belonging to a single cluster can be combined (aggregated). The node with highest available network resources is selected as a cluster head.

Communication based. The routing protocols based on communication are classified as unicast, multicast and